

Software Testing

Module 6

Module-End Assignment:

“AI in Software Testing”

Q1: Configure the automation tool and set up a new project for web testing.

Steps:

1. Install and launch Katalon Studio or log in to BrowserStack Automate.
2. Create a new web automation project.
3. Configure the target browser (e.g., Chrome, Firefox).
4. Set the base URL of any sample web application (for example:
 <https://the-internet.herokuapp.com/login>).
5. Expected Output:
Project initialized successfully and browser configuration verified.

Katalon Studio:

Steps Performed:

1. Install and open Katalon Studio.
2. Create a new project by selecting:
File → New → Project.
3. Enter the project name: Web automation.
4. Select the project location and click OK/Create.
5. Use the sample application URL: <https://the-internet.herokuapp.com/login>
6. Project initialized successfully and browser configuration verified.

Q2: Use the record-and-playback feature to automate a basic web flow.

Steps:

1. Start the Record session in Katalon or use BrowserStack's recorder.
2. Perform actions like:
 - Open the URL
 - Enter sample data in the login form (Name, Email, Password)
 - Click on Login
3. Stop the recording and save the test case.
4. Run the recorded test to verify successful execution.

Expected Output:

Test case runs successfully, registering the user and confirming form submission.

Katalon Studio:

Steps Performed:

1. Open Katalon Studio.
2. Create a new test case.
3. Click the Record Web option.
4. Select the Chrome browser.
5. Enter the application URL.
6. Start recording.
7. Perform the required actions on the application:
 - Enter the username.
 - Enter the password.
 - Click the Login button.
8. Stop the recording.
9. Save the recorded test case.
10. Run the saved test case using the Run option.
11. Verify that all recorded steps execute successfully.

Actual Result:

The browser performed the recorded actions automatically, and the test flow was executed successfully.

Status: PASS

Q3: Verify that the automation test includes key validations.

Steps:

1. Add checkpoints or assertions to confirm:
 - Page title is correct.
 - Registration success or validation message appears.
2. Re-run the test to validate these assertions.

Expected Output:

All verification steps pass successfully during test execution.

Katalon Studio:

Steps Performed:

1. Open the recorded test case.
2. Identify the page title or expected success message.
3. Add a verification step to check whether the page title is correct.
4. Add another verification step to check whether a success or validation message appears.
5. Save the test case.
6. Execute the test again.
7. Check the execution results.

Q4: Create and export an HTML or PDF report of test execution.

Steps:

1. Execute the saved test case or suite.
2. Generate an execution report within the tool.
3. Export the report in HTML/PDF format.
4. Include screenshots for passed or failed steps.

Expected Output:

Execution report generated with detailed pass/fail status and evidence.

Katalon Studio:

Steps Performed:

1. Execute the saved test case or test suite.
2. Wait until the execution is completed.
3. Open the execution results in Katalon Studio.
4. Review the status of each test step.
5. Check the passed and failed results.
6. Generate the execution report.
7. Export or save the report in HTML or PDF format, depending on the available reporting option.

Actual Result:

The execution results were successfully generated and documented in the report.

Status: PASS

Q5: Run the same recorded test case across multiple browsers using BrowserStack or local browser setup.

Steps:

1. Configure additional browser options (Chrome, Edge, Firefox).
2. Execute the test across each environment.
3. Compare execution results.

Expected Output:

Consistent test results across browsers with screenshots or logs for reference.

Katalon Studio:

Steps Performed:

1. Create a Test Case and run it.
2. Create a Test Suite.
 - Go to Test Suites → New Test Suite.
 - Name the Test Suite.
 - Add the already created Test Case in this suite.
3. Create the Test Suite Collection.
 - Go to Test Suite Collections → New Test Suite Collection.
 - Name the Test Suite Collection.
 - Add the already created Test Suite in this.
4. Add 3 execution entries.
 - In the collection, add the same Test Suite three times.
 - Configure each entry with a different browser:
 - Entry 1 → Chrome
 - Entry 2 → Microsoft Edge
 - Entry 3 → Firefox
5. Enable parallel execution.
 - In the Test Suite Collection, set the execution mode to Parallel.
 - This makes the three browser executions run at the same time.

6. Run.

- Select Run.
- Katalon should launch Chrome + Edge + Firefox simultaneously.

Actual Result:

The test case was executed on Chrome, Edge, and Firefox. The same test flow worked successfully on all configured browsers.

Status: PASS